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“Really bizarre” quantum discovery defies the rules of physics

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Source: University of Michigan

Summary: Researchers have discovered quantum oscillations inside an insulating material, overturning long-held assumptions. Their work at the National Magnetic Field Laboratory suggests that the effect originates in the material’s bulk rather than its surface. The finding points toward a “new duality” in materials science—where compounds may behave as both metals and insulators—offering a fascinating puzzle for future research.

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FULL STORY



An international team has found that quantum oscillations can arise from the bulk of an insulator, not just its surface. The discovery exposes a new frontier where materials can act like metals under extreme magnetic fields. Credit: AI/ScienceDaily.com

Lu Li, a physicist who studies advanced materials, knows that people often want to hear how his research could lead to new technologies or practical breakthroughs. But sometimes, what he uncovers is so unusual that its value lies purely in revealing how strange the universe can be.

Working with an international team of scientists, Li has made one of those discoveries, recently described in *Physical Review Letters*.

"I would love to claim that there's a great application, but my work keeps pushing that dream further away," said Li, a professor of physics at the University of Michigan. "But what we've found is still really bizarre and exciting."

Quantum Oscillations: When Electrons Act Like Springs

Supported by the U.S. National Science Foundation and the U.S. Department of Energy, the research focuses on a puzzling effect called quantum oscillations. In metals, these oscillations occur when electrons behave like tiny springs, vibrating in response to magnetic fields. By changing the magnetic field's strength, scientists can alter how quickly these "electron springs" move.

In recent years, however, researchers have discovered the same quantum oscillations in insulators - materials that should not conduct electricity or heat. That revelation has left scientists debating whether the effect originates only on the surface of these materials or deep within their interior

(known as the bulk).

Searching for Answers Inside the Material

If the oscillations came from the surface, that would be particularly exciting for potential technologies. Materials called topological insulators, which conduct electricity on their surfaces while remaining insulating inside, are already being studied for new kinds of electronic, optical, and quantum devices.

To explore the mystery, Li and his collaborators turned to the National Magnetic Field Laboratory, home to the most powerful magnets in the world. Their experiments revealed that the oscillations were not just a surface effect. Instead, they came from the bulk of the material itself.

"I wish I knew what to do with that, but at this stage we have no idea," Li admitted. "What we have right now is experimental evidence of a remarkable phenomenon, we've recorded it and, hopefully, at some point, we'll realize how to use it."

A Global Collaboration and a Clear Result

The study involved more than a dozen scientists from six institutions in the United States and Japan, including research fellow Kuan-Wen Chen and graduate students Yuan Zhu, Guoxin Zheng, Dechen Zhang, Aaron Chan, and Kaila Jenkins from the University of Michigan.

"For years, scientists have pursued the answer to a fundamental question about the carrier origin in this exotic insulator: Is it from the bulk or the surface, intrinsic or extrinsic?" said Chen. "We are excited to provide clear evidence that it is bulk and intrinsic."

A "New Duality" in Physics

Li describes the finding as part of what he calls a "new duality." The original, or "old," duality in physics emerged more than a century ago when scientists realized that light and matter can act as both waves and particles. That discovery transformed physics and led to technologies such as solar cells and electron microscopes.

The new duality, Li says, involves materials that can behave as both conductors and insulators. His team explored this idea using a compound called ytterbium boride (YbB₁₂) inside a magnetic field so powerful that it reached 35 Tesla -- about 35 times stronger than the field inside a hospital MRI machine.

"Effectively, we're showing that this naive picture where we envisioned a surface with good conduction that's feasible to use in electronics is completely wrong," Li explained. "It's the whole compound that behaves like a metal even though it's an insulator."

Unlocking the Mystery of a "Crazy Metal"

Although this "metal-like" behavior only appears under extreme magnetic conditions, the finding raises new questions about how materials behave at the quantum level.

"Confirming that the oscillations are bulk and intrinsic is exciting," said Zhu. "We don't yet know what kind of neutral particles are responsible for the observation. We hope our findings motivate further experiments and theoretical work."

The project received additional support from the Institute for Complex Adaptive Matter, the Gordon and Betty Moore Foundation, the Japan Society for the Promotion of Science, and the Japan Science and Technology Agency.

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Journal Reference:

1. Kuan-Wen Chen, Yuan Zhu, Danilo Ratkovski, Guoxin Zheng, Dechen Zhang, Aaron Chan, Kaila Jenkins, Joanna Blawat, Tomoya Asaba, Fumitoshi Iga, Chandra M. Varma, Yuji Matsuda, John Singleton, Alimamy F. Bangura, Lu Li. **Quantum Oscillations in the Heat Capacity of Kondo Insulator YbB12**. *Physical Review Letters*, 2025; 135 (15) DOI: [10.1103/ms3x-pjsk](https://doi.org/10.1103/ms3x-pjsk)

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